LOS ANGELES COUNTY SHERIFF'S DEPARTMENT

Scientific Services Bureau — Crime Laboratory

Forensic Biology / Toxicology Section

CERTIFICATE OF ANALYSIS
Blood Alcohol Concentration — Evidentiary Report

1800 Paseo Rancho Castilla, Los Angeles, CA 90032 • ANAB ISO/IEC 17025 Accredited Forensic Testing Laboratory

CASE AND SPECIMEN INFORMATION

Laboratory Case No.: LASD-SSB-25-0442117
Submitting Agency: Rivergate Police Department
Agency Report No.: RPD TC-2025-1018
Related Criminal Case: People v. Raymond T. Keeler — LASC Norwalk No. NA123456-01
Charge Reference: Cal. Veh. Code § 23153(a) — DUI Causing Injury
Subject (Suspect): KEELER, Raymond T.
Subject DOB: 04/11/1979
Subject Address of Record: 1837 Alameda Street, Apt 4C, Rivergate, CA 91766
Collection Date: 10/18/2025
Collection Time: 03:10 (approx.)
Collection Location: Rivergate Regional Trauma Center — Trauma Bay 2
Collected By: Certified phlebotomist under RPD DUI blood-draw protocol; kit lot LASD-BK-2411
Specimen Description: Whole blood — two (2) gray-top vacutainer tubes, sodium fluoride / potassium oxalate preservative; sealed
Chain of Custody Document: LASD-SSB Form COC-25-0442117 (attached — reference the Blood-Draw Chain-of-Custody
Received at Laboratory: 10/20/2025, 08:47 hrs; seals intact; refrigerated storage upon intake
Analyst: Alan Whitford, Criminalist III
Technical Reviewer: Loretta M. Ainsley, Criminalist IV — Toxicology Discipline Lead
Report Date: 11/07/2025

ANALYTICAL METHOD

Analysis was performed pursuant to LASD-SSB Toxicology SOP TOX-102 (rev. 7), "Determination of Ethanol in Whole Blood by Headspace Gas Chromatography with Flame Ionization Detection."

• **Technique:** Automated static headspace sampling with dual-column gas chromatography and flame ionization detection (HS-GC-FID).

• **Instrument:** Agilent 7890B GC / G1888 headspace autosampler; primary column DB-ALC1, confirmatory column DB-ALC2 (parallel dual-column analysis).

• **Internal Standard:** n-propanol.

• **Sample Handling:** Duplicate aliquots taken from each of the two submitted tubes. Each aliquot analyzed on both columns. Quantitation reported as the mean of duplicate determinations; inter-column agreement within ±5 % of the mean, meeting SOP acceptance criteria.

• **Volatiles Screen:** Simultaneous screen for methanol, acetone, and isopropanol from the same headspace injection.

CALIBRATION AND QUALITY CONTROL

Calibration and control materials in the analytical batch were traceable to NIST-referenced standards.

Calibration Standards (aqueous ethanol, Cerilliant / NIST-traceable, Lot AE-25-0918):

Level	Nominal Value (g/dL)	Observed (g/dL)
Cal-1	0.020	0.020
Cal-2	0.050	0.050
Cal-3	0.080	0.081
Cal-4	0.100	0.101
Cal-5	0.200	0.199
Cal-6	0.400	0.398

Calibration curve: linear, weighted 1/x; r^2 = 0.9999 (primary column) / 0.9999 (confirmatory column). Curve verified prior to and following the batch.

Quality Control Materials (whole-blood matrix controls, UTAK Laboratories):

Control	Target (g/dL)	Observed (g/dL)	Acceptance	Result
Negative control (blank blood)	< 0.010	ND	< 0.010	Pass
Low QC	0.040	0.040	±10 %	Pass
Mid QC	0.100	0.101	±10 %	Pass
High QC	0.300	0.298	±10 %	Pass
Method blank	0.000	ND	< 0.010	Pass
Reagent blank	0.000	ND	< 0.010	Pass

All controls, calibrators, and blanks were within established acceptance limits for the analytical batch containing this specimen. The batch was accepted by the analyst and the technical reviewer.

RESULTS

Analyte	Matrix	Result	Units	Method
Ethanol — primary column	Blood (whole)	**0.14**	% w/v (g/dL)	HS-GC-FID
Ethanol — confirmatory column	Blood (whole)	**0.14**	% w/v (g/dL)	HS-GC-FID
Methanol	Blood (whole)	None Detected	—	HS-GC-FID
Acetone	Blood (whole)	None Detected	—	HS-GC-FID
Isopropanol	Blood (whole)	None Detected	—	HS-GC-FID

Reported Blood Alcohol Concentration: 0.14 % w/v (0.14 g/dL).

The statutory *per se* threshold under California Vehicle Code § 23152(b) is 0.08 % w/v. The reported value exceeds that threshold. The commercial driver *per se* threshold under Vehicle Code § 23152(d) is 0.04 % w/v; the reported value likewise exceeds that threshold.

Reporting limit for ethanol under this method: 0.010 g/dL. Expanded measurement uncertainty at the reported concentration: ±0.005 g/dL (k = 2, approximate 95 % confidence).

Remaining specimen and the second unopened tube have been retained in secured, refrigerated laboratory storage under the retention schedule set forth in LASD-SSB Evidence Retention Policy EV-04.

CHAIN OF CUSTODY

An unbroken chain of custody accompanies this report as attachment the Blood-Draw Chain-of-Custody (LASD-SSB Form COC-25-0442117), documenting the specimen from the point of collection at Rivergate Regional Trauma Center on 10/18/2025 at 03:10, through law-enforcement transport to the Los Angeles County Sheriff's Department Scientific Services Bureau evidence intake, refrigerated storage, analytical handling, and return to secured storage. All seals were intact upon each transfer. All transfers are initialed and dated by the transferring and receiving custodians.

LABORATORY CERTIFICATION STATEMENT

The Los Angeles County Sheriff's Department Scientific Services Bureau — Crime Laboratory is accredited by the ANSI National Accreditation Board (ANAB) to ISO/IEC 17025:2017 for forensic testing (Certificate No. AT-1653, Toxicology discipline). This certificate of analysis has been prepared and reported in accordance with the Laboratory's Quality Management System and Toxicology Discipline SOPs in effect on the date of analysis.

I, Alan Whitford, hereby certify that the analyses reported herein were performed by me or under my direct supervision using validated methods; that the calibrators, controls, blanks, and quality-assurance requirements applicable to this determination were met; that the specimen was in the same condition on receipt as reflected in the chain-of-custody records; and that the results set forth above are true and correct to the best of my knowledge, information, and belief.

I declare under penalty of perjury under the laws of the State of California that the foregoing is true and correct.

Executed at Los Angeles, California, on 11/07/2025.

Signed: ____

Alan Whitford, Criminalist III

Los Angeles County Sheriff's Department

Scientific Services Bureau — Crime Laboratory

Toxicology Section

Analyst Credentials:

• B.S., Forensic Science, California State University Los Angeles (2008)
• ABC-GKE (American Board of Criminalistics — General Knowledge Examination), certified 2011
• ABC Diplomate — Forensic Alcohol / Toxicology Specialty, certified 2014; recertification current
• California Association of Criminalists — member in good standing
• SWGTOX (Scientific Working Group for Forensic Toxicology) methods training, 2013 and continuing education current
• Court-qualified as an expert in the analysis of ethanol in biological specimens and forensic toxicology in the Superior Courts of the State of California, Counties of Los Angeles, Orange, San Bernardino, and Ventura
• Fourteen (14) years' experience in forensic toxicology at LASD-SSB

TECHNICAL REVIEW CONCURRENCE

The analytical data, quality-assurance records, and chain-of-custody documentation supporting this report have been independently reviewed by a second qualified analyst and found to be complete, accurate, and in conformance with Laboratory SOPs.

Signed: ____

Loretta M. Ainsley, Criminalist IV

Toxicology Discipline Lead

Date of Technical Review: 11/07/2025

End of Report — LASD-SSB Case No. 25-0442117 — 1 specimen, 2 tubes, single subject (KEELER, Raymond T.). Distribution: Rivergate Police Department (submitting agency, ref. RPD TC-2025-1018); Los Angeles County District Attorney's Office (Norwalk Branch, People v. Keeler, NA123456-01); Laboratory case file.